

NCIE-010 Skeletal v0.1 → Full v1.0

Expansion Verification Report

Evidence, Provenance & Audit Architecture — IN DEVELOPMENT / FOR HUMAN REVIEW OF FULL EXPANSION

1. Scope Verification

NCIE-010 v1.0 is a controlled full expansion of the approved NCIE-010 Skeletal v0.1: every approved chapter purpose, Salient Things to Note, principal component, upstream dependency/authority and Human Review focus is elaborated into Evidence lifecycle, chain-of-custody, provenance and audit architecture, without inventing evidence-management, graph, audit, ledger, archival, hashing, storage, SIEM, database or workflow products, without prescribing specific hash/cryptographic algorithms, and without inventing retention periods, custody-attestation thresholds, admissibility thresholds, audit-review periods, assurance frequencies or evidentiary confidence cutoffs.

2. 29-Chapter Integrity Verification

A programmatic scan of the final document confirms exactly 29 sequential 'NCIE-010 — Chapter N Expansion' labels, N = 1..29, each present exactly once, in the approved order, with no chapter added, removed, merged, split or renamed relative to the approved Skeletal v0.1 structure.

3. Salient Things to Note Hard-Gate Verification

Per directive items 5 and 70 — the latter explicitly naming the NCIE-009 Chapter-27 omission as the failure mode not to repeat — every chapter must retain an explicit '<N>.0 Salient Things to Note' subsection. This build enforced the requirement structurally rather than only procedurally: the chapter-rendering helper (lib.py's add_chapter function) raises a build-time assertion error if a chapter's salient content is omitted, so the document cannot be assembled with a missing subsection in the first place. A programmatic scan of the final assembled document, rerun independently rather than inferred from the build script succeeding, confirms an explicit Salient Things to Note heading present for all 29 chapters (N = 1 through 29), with zero missing.

4. Skeleton Fidelity Verification

Each of the 29 chapters was authored directly from its approved skeletal Chapter Purpose, Salient Things to Note, Principal Components, Upstream Dependencies/Authority and Human Review Focus (see the accompanying Change Register for the per-chapter mapping). No approved chapter concept was omitted, and none was conceptually inverted.

5. No Product, Algorithm or Threshold Invention

A full-text scan for named evidence-management, graph, audit, ledger, archival, hashing, storage, SIEM, database or workflow products found none invented (directive item 8) — Chapter 6 and Chapter 11 explicitly defer product/algorithm selection. A separate scan for specific hash/cryptographic algorithms or signature schemes found none prescribed (directive item 9) — Chapter 6 states required integrity properties only. A third scan for retention periods, custody-attestation thresholds, admissibility thresholds, audit-review periods, assurance frequencies or evidentiary confidence cutoffs found none invented (directive item 10) — every such matter is instead recorded as a named Human Review item (e.g., HR10-6-1 for integrity assurance, HR10-24-1 for retention periods, HR10-27-1 for evidentiary standards).

6. Governing Evidence/Provenance/Audit Invariants Verification

The 19 invariants proposed in the approved Skeletal v0.1 ('Proposed Governing Invariants') are restated unchanged in front matter §0.3, and are additionally cross-referenced — not restated — against the 12 NCIE-008 governance invariants, the 16 NCIE-009 security invariants, and other upstream invariants already binding from NCIE-005/006/007. A full-text scan confirms every invariant is both stated once in §0.3 and applied by name in at least one operative chapter (e.g., SOURCE DATA ≠ EVIDENCE in Ch.4; HASH/INTEGRITY METADATA ≠ TRUTH OF CONTENT in Ch.6; CONTRADICTION ≠ ERROR TO BE SILENTLY REMOVED in Ch.11; LINEAGE ≠ CAUSALITY in Ch.18; DECISION-TIME RECONSTRUCTION ≠ CURRENT-STATE REWRITE in Ch.16).

7. Evidence/Governance/Security Architecture Boundary Verification

Chapter 1 Table 1 fixes what NCIE-010 does not own (product requirements, architecture, canonical data, engineering baseline, ARGUS behavior, Memory/Context architecture, orchestration, institutional governance/RACI, and security/IAM enforcement — all NCIE-001–009's). Every consequential Evidence/audit control in Chapters 4–27 is expressed as enforcement of a named NCIE-008 decision-right class and/or an NCIE-009 security control with the accountable holder marked 'to be assigned,' never as NCIE-010 assigning institutional authority or security mechanisms of its own (directive item 7).

8. Contradiction-Preservation Audit

A full-text scan (directive item 59) for wording that would permit contradictory Evidence to be silently removed, overwritten or summarized away found none — Chapter 10 §10.1, Chapter 11 §11.1–11.2 and the Evidence Relationship

Matrix (Table 13) each explicitly require contradictions to persist through relationships, summaries, AI transformations and Context compression, with resolution requiring an explicit resolution event rather than disappearance.

9. Evidence-Status Audit

A full-text scan (directive item 60) for wording that would make source data, Memory, AI/Agent output, VPF PASS, annotation or search result automatically become Evidence found none — Chapter 2 §2.2, Chapter 4 §4.1, Chapter 12 §12.1, Chapter 14 §14.1 and Chapter 25 §25.1 each independently restate the applicable non-equivalence (SOURCE DATA ≠ EVIDENCE, AI/AGENT OUTPUT ≠ EVIDENCE, VPF PASS ≠ EVIDENCE, ANNOTATION ≠ ALTERATION OF SOURCE EVIDENCE, SEARCH RESULT ≠ EVIDENCE STATE).

10. Finding/Decision Audit

A full-text scan (directive item 61) for wording that would allow Evidence, AI output, VPF state or provenance alone to become an institutional Finding/Decision found none — Chapter 2 §2.2, Chapter 17 §17.1 and Chapter 21 §21.1 each confirm EVIDENCE SUPPORT ≠ INSTITUTIONAL DECISION, FINDING ≠ DECISION, and that Governance Decision records remain distinct from AI/VPF states and execution receipts.

11. Historical-Integrity Audit

Verified (directive item 62) that correction (Ch.8), supersession (Ch.8), invalidation/withdrawal (Ch.9) and recovery (Ch.26) each preserve historical state and Decision-Time interpretability: corrections create new versions rather than overwriting (CORRECTION ≠ HISTORICAL ERASURE), invalidation does not delete the historical record (INVALIDATION ≠ DELETION OF HISTORICAL RECORD), and Chapter 16's Decision-Time Reconstruction Matrix (Table 20) fixes exactly what historical state each artifact class needs at a cutoff with explicit unavailable-state handling rather than silent substitution of current data.

12. Security/Disclosure Audit

Verified (directive item 63) that Evidence/provenance/audit retrieval and graph connectivity cannot bypass NCIE-009 authorization, classification, DLP, purpose or protected-identity controls: Chapter 23 confirms PROVENANCE ≠ AUTHORITY TO DISCLOSE and requires classification inheritance into provenance/audit records; Chapter 25 confirms SEARCH RESULT ≠ EVIDENCE STATE and that semantic retrieval is not authority to disclose; Chapter 20 confirms audit itself never grants disclosure authority independent of a live NCIE-009 authorization check.

13. Evidence Lifecycle, Custody and Relationship Matrix Verification

Chapter 4 Table 5 (Evidence Lifecycle Matrix) traces provenance and audit obligations across 9 lifecycle stages from source registration to disposition, each citing its owning chapter. Chapter 7 Table 9 (Chain-of-Custody Matrix) fixes handler identity, purpose, transfer/access type, integrity check and state transition for 6 custody-event classes without inventing institutional actors beyond the abstract handler role. Chapter 10 Table 13 (Evidence Relationship Matrix) fixes allowed endpoints and a semantic caution for all 6 relationship types (supports, contradicts, corroborates, duplicates, derives-from, related-to).

14. Provenance-versus-Audit and Decision-Time Matrix Verification

Chapter 19 Table 24 explicitly compares provenance, audit and telemetry by purpose, object, authority, retention interface, security sensitivity and Decision-Time use (directive item 53), so the three record classes cannot be conflated anywhere in this document. Chapter 16 Table 20 (directive item 54) identifies the historical state needed for Evidence, transformations, provenance, audit, Tool outcomes, AI artifacts, Findings and Decisions at a cutoff, with explicit unavailable-state handling for each artifact class.

15. Evidence Defensibility Package Verification

Chapter 27 §27.3 (directive item 55) defines the seven-element logical package required to defend a material Finding/Decision — source identity, Evidence versions, custody, transformations, contradictions/gaps, governance approvals, and audit references — explicitly stating that no single element is sufficient alone and that assembling the package does not itself constitute institutional review, which remains NCIE-008's authority. No institutional admissibility rule is invented in stating this package (directive item 64).

16. AI/Agent/ARGUS and Human Contribution Provenance Verification

Chapter 14 confirms AI/Agent/ARGUS output remains AI-derived unless an independent governed process establishes it as Evidence, with a mandatory output-classification field on every AI-derived artifact. Chapter 15 confirms human origin alone does not make a statement Evidence or a Finding, with a mandatory contribution-type field on every human-attributed record.

17. Test-Status Integrity Verification

Chapter 27 Table 33 specifies 8 test classes (chain-of-custody, tamper detection, version-history integrity, contradiction preservation, Decision-Time reconstruction, audit completeness, authorization-audit, recovery), each traced to the chapter(s) it verifies. Every test is marked TEST SPECIFIED only; a full-text scan confirms no test anywhere in this document is marked TEST EXECUTED, TEST PASSED, or claims IMPLEMENTATION EVIDENCE AVAILABLE (directive items 58 and 66) — detailed test-case authoring and execution are explicitly deferred to NCIE-016.

18. No Fabricated Admissibility Law or Technology

A full-text scan for statutory evidentiary standards, court admissibility rules, or regulatory retention obligations found none invented (directive item 64). A separate scan for ledger/blockchain, immutable database, graph database, WORM storage, or specific hashing/signing technology found none invented merely to complete the architecture (directive item 65) — every such matter is stated as a required property (e.g., tamper-evidence, version lineage) rather than a named product.

19. Human Review Register Mechanical Reconciliation (Correct From First Build)

Chapter 28's register was recomputed programmatically by scanning the Human Review Focus bullet text already rendered in Chapters 1-27, plus HR10-28-1 (Chapter 28's own item) and HR10-29-1 (Chapter 29's own item) — both defined once and included in the register-construction list from the start of the build, a design choice made specifically to avoid repeating the register-reconciliation defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Result: 29 total items, 22 Blocking / 7 Non-Blocking, 0 duplicate Review IDs, 0 orphan Human Review items. Classification breakdown: 25 Institutional Confirmation Required, 2 Security/Sovereignty Confirmation Required (Chapters 20, 23), 1 Legal/Policy Confirmation Required (Chapter 24 retention/legal hold), 1 Resolved by Upstream Baseline (Chapter 29).

20. Upstream Human Review Deduplication (Directive Item 57)

HR10-1-1 (confirm NCIE-010 acceptance and audit-overlap resolution) restates the identical institutional question NCIE-008 already opened as HR8-10-1 (acceptance authorities for each NCIE-series document and chapter-level sign-off); Chapter 28 Table 36 cross-references HR8-10-1 rather than treating HR10-1-1 as an independent open question. Every other NCIE-010 Human Review item was individually checked against the NCIE-008/009 Human Review Registers and found to concern an Evidence/provenance/audit-specific decision that NCIE-008/009 do not themselves resolve — no other duplicate was found.

21. Blocking Scope Discipline Verification

Each Blocking item in the Chapter 28 register states precisely which control default, threshold, ownership assignment or acceptance authority it blocks, rather than implying the whole document or the whole approved upstream chain is blocked. Non-Blocking items (Chapters 2, 10, 12, 15, 18, 21, and the resolved Chapter 29 item) are explicitly marked as not gating the corresponding chapter's Evidence/provenance/audit operation in the interim.

22. Traceability and Downstream Non-Preemption Verification

Chapter 29 Table 37 traces all 29 chapters to their primary upstream authority (approved NCIE-001-009 and/or the approved NCIE-010 Skeletal v0.1); no chapter states an Evidence/provenance/audit requirement without an identifiable source. Table 38 confirms this document hands over interface contracts to NCIE-011, logical retention/disposition requirements to NCIE-014, operational responsibilities to NCIE-015, and the 8 verification/assurance test classes to NCIE-016 — without specifying their technical implementation, consistent with directive item 20 (downstream non-preemption).

23. No Implementation Claims

A full-text scan confirms no chain-of-custody, audit-completeness, tamper-detection, historical-reconstruction or defensibility control anywhere in this document is claimed implemented, deployed, tested or production-approved merely because this v1.0 specifies it (directive item 66) — every RACI/control-ownership table marks its holder 'to be assigned,' and Chapter 27 explicitly marks every test SPECIFIED, not executed.

24. Version, Document-Control and Running-Header Verification

Cover and Document Governance table confirmed reading: Edition 'In development — Version 1.0 (Full Expansion of Approved Skeletal v0.1)'; Requirements/Architecture/Canonical-data/Engineering/ARGUS-behavioral/Memory-Context/Orchestration/Governance/Security authority rows all read their approved current versions (NCIE-001 v1.3 through NCIE-009 v1.0); a 'Prior baseline' row records the approved Skeletal v0.1 as historically preserved; Status reads IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION. The running header — rendered from a copied NCIE-009 v1.0 template to inherit this series' visual styling, including the logo, centered cover text, page break before Document Governance, and non-splitting governance-table rows already corrected in that series — was corrected to 'NCIE-010 | EVIDENCE, PROVENANCE & AUDIT ARCHITECTURE | VERSION 1.0' and the footer corrected from 'CONTROLLED SECURITY DOCUMENT' to 'CONTROLLED EVIDENCE DOCUMENT' and from 'NCIE-009' to 'NCIE-010' before this document was rendered to PDF.

25. Full-Document Cross-Reference Scan

Every occurrence of 'NCIE-009' in the final document was individually inspected (32 occurrences outside Upstream/Trace blocks). Each was confirmed to be a legitimate security-control citation (e.g., 'NCIE-009 artifact-security owner', 'Accessing principal (NCIE-009 authenticated)') consistent with directive item 7's requirement that NCIE-010 consume, not redefine, NCIE-009 security controls — none is a leftover donor-template branding error. The cover, header, footer and Document Governance 'Document' row were separately confirmed to read NCIE-010, not NCIE-009.

26. Character-Encoding Integrity Check

Following the same build pipeline used for the whole NCIE series (UTF-8 source fragments executed via a Python `exec()` call with an explicit `encoding='utf-8'` argument, a fix first required during the NCIE-008 build), a full-document scan of the final NCIE-010 document confirmed zero mojibake occurrences across all 598 paragraphs and 39 tables.

27. Visual and Structural Verification

The complete document was opened in Microsoft Word, its Table of Contents field (a live Word field, not static text) was updated and confirmed to populate with correct chapter/section titles and real page numbers, repaginated to 52 pages, saved, and rendered to PDF. Independent text extraction from the rendered PDF confirms the cover occupies page 1 alone and Document Governance begins on page 2 (the manual page break), and that the Document Governance table's rows do not split across the page 2/3 boundary (the `cantSplit` fix already established for this series). No table clipping, overlapping text, or broken table was observed across the five consolidated matrices, the Chapter 28 Human Review Register tables, or the Chapter 29 traceability/handover tables.

28. Final Human Review Extraction (Authoritative)

Performed directly against the final specification file by independently scanning every paragraph and table cell for the pattern `HR10-<chapter>-<n>`: TOTAL IDS FOUND = 29. UNIQUE IDS = 29. DUPLICATES = 0. OPEN = 28. RESOLVED = 1 (HR10-29-1, Resolved by Upstream Baseline). BLOCKING = 22. NON-BLOCKING = 7. CLASSIFICATION TOTALS: Institutional Confirmation Required = 25; Security/Sovereignty Confirmation Required = 2; Legal/Policy Confirmation Required = 1; Resolved by Upstream Baseline = 1. ORPHAN IDS = 0 — every one of the 29 unique IDs found anywhere in the document also appears as a row in the Chapter 28 Complete Register table, which itself contains exactly 29 rows with no duplicate Review ID.

29. Three-Artifact Consistency Check

The NCIE-010 v1.0 main document, this Verification Report and the accompanying Full-Expansion Change Register agree exactly on: Source Version v0.1 (Skeletal) → Target Version v1.0 (Full); Authority chain approved NCIE-001 v1.3 through approved NCIE-009 v1.0, plus the approved NCIE-010 Skeletal v0.1; Status IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION (not Approved); 29 chapters preserved in order (0 missing, 0 duplicate, 0 added, 0 unauthorized renaming); Salient Things to Note hard gate = 29/29 explicit subsections present; Human Review Register TOTAL = 29, BLOCKING = 22, NON-BLOCKING = 7, classification totals as in §28 above; orphan Human Review IDs = 0; duplicate Human Review IDs = 0; verification/assurance test-class status = 8 tests specified, none executed or passed; Evidence/provenance/audit invariants = 19 NCIE-010-specific plus cross-referenced upstream invariants; downstream handover = NCIE-011/014/015/016 each receive a defined interface without technical preemption. No mismatch was found between the three artifacts.

30. Final Readiness Disposition

NCIE-010 — Evidence, Provenance & Audit Architecture v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed control default, threshold, ownership assignment and product/algorithm choice deferred in Chapters 1-27 remains a proposal pending resolution of the Chapter 28 Human Review Register; every §27.1 verification/assurance test remains SPECIFIED, not executed, pending NCIE-016. This document is NOT marked Approved.